MILITARY SPECIFICATION

AMPLIFIER MARK 140 MODS O, 1, AND 2

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1, SCOPE

1.1 Scope. - This specification establishes the requirements for the procurement of Amplifier Mk 140 Mods O, 1, and 2, Bureau of Ordnance LDs 426721, 426716, and 426723, respectively. Amplifier Mk 140 is an electronic fire control instrument which amplifies synchronous torpedo input orders to the power level required to drive Mk 7 or Mk 8 torpedo control mechanism servomotors. Amplifier Mk 140 Mods O, 1, and 2 is of one type and one class.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications or such portions thereof as designated herein of the issue in effect on the date of invitation for bids form a part of this specification.

SPECIFICATIONS

MILITARY

MIL-F-1887OA	Fire Control, Optical and Associated Equipment Intended for Shipboard Use; The General Specification for Design and Procurement of
MIL-P-116	Preservation, Methods of
MIL-P-17555	Preparation of Delivery of Electronic Equipment
MIL-T-21719	Test Set Mk 339 Mod O

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

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MIL-STD-108	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-129	Markings for Shipment and Storage

DRAWINGS

NAVY DEPARTMENT

BUREAU OF ORDNANCE

BUORD LD 426716	Amplifier Mk 140 Mod 1, List of Drawings, Subassemblies, Parts, and Specifications
BUORD LD 426717	Gyro Angle Function Amplifier, List of Drawings, Subassemblies, Parts, and Specifications
BUORD LD 426718	Running Depth Function Amplifier, List of Drawings, Subassembliess, Parts? and Specifications
BUORD LD 426719	Enabling Run Function Amplifier, List of Drawings, Subassemblies, Parts, and Specifications
BUORD LD 426721	Amplifier Mk 140 Mod 0, List of Drawings, Subassemblies, Parts, and Specifications
BUORD LD 426723	Amplifier Mk 140 Mod 2, List of Drawings, Subassemblies, Parts, and Specifications
DRAWING NO. 876118	Electrical Settings for Torpedoes, Schematic Diagram
DRAWING NO. 1430904	Filter, Band-pass, 400-Cycle
DRAWING NO. 1485523	Outline and Installation, Amplifier Mk 140 Mod 1
DRAWING NO. 1485524	Outline and Installation, Amplifier Mk 140 Mod 2
DRAWING NO. 1485525	Outline and Installation, Amplifier Mk 140 Mod 0

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DRAWING NO. 1485531	Shipping Case, Amplifier Mk 140 Mod O
DRAWING NO. 1485532	Shipping Case, Amplifier Mk 140 Mod 1
DRAWING NO. 1485533	Shipping Case," Amplifier Mk l4O Mod 2
DRAWING NO. 1287297	Amplifier Assembly, Magnetic

PUBLICATIONS

NAVY DEPARTMENT

BUREAU OF ORDNANCE

OP 400	General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment
OP 2555	Amplifier Mk 140 Mods O, 1, and 2, and Test Set Mk 339 Mod O, Description, Operation, and Maintenance
NAVORDOD 12405	Amplifier Mk 140 Mods O, 1, and 2, Factory Acceptance Test
NAVORD OD 9154	Electrical Settings for Torpedoes; Specifications for

(Copies of specifications, . standards, drawings, and publications r e - quested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28 and 1950 Supplement	Screw Thread Standards for the Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. G.)

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3. REQUIREMENTS

3.1 Description. - Amplifier Mk 140 is an electronic fire control instrument which amplifies synchronous torpedo input orders to the power level required to drive Mk 7 or Mk 8 torpedo control mechanism servomotors. Amplifier Mk 140 also develops signals to operate synchronism indicators. The instrument is used in submarine and surface vessel fire control systems where input orders are electrically set into torpedoes.

Amplifier Mk 140 consists of individual servo amplifiers arranged in groups of-three (which are called channels). One servo amplifier in each channel amplifies gyro angle order inputs, another amplifies running depth order inputs, and the third amplifies enabling run order inputs. The differ - ence between the three mods of Amplifier Mk 140 is in the number of channels each contains and, therefore, the number of torpedoes each -controls. Mod O has four channels, Mod 1 has three, and Mod 2 has two channels.

The individual servo amplifiers are plug-in units which contain 24-volt DC relays to permit compatible operation with the operational modes of various types of torpedoes. All servo amplifiers plug into a base assembly containing connector jacks for the amplifiers and terminal boards for connecting ship wiring. Each servo amplifier is guide-pin-indexed to prevent inter change with another servo amplifier of a different function. A gasket-sealed, spraytight cover encloses the servo amplifiers and fastens to the base assembly to make all mods of Amplifier Mk 140 of spraytight construction.

Operation of the equipment is completely automatic and does not require the attention of an operator. A complete description of the instrument is included in OP 2555.

3.2 General requirements

3. 2.1 When specifically required by the contract or order, pre-production samples of Amplifier Mk 140 Mods O, l, and 2 shall be furnished for tests as specified in section 4 herein for the purpose of determining that the production item meets the requirements of this specification.

3. 2.2 Materials. - Unless otherwise specified in the contract; requisition, or order, all materials and purchased parts entering into the final assembly of Amplifier Mk 140 shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in BUORD LDs 426716, 426717, 426718, 426719, 4.26721, and 426723.

3. 2.3 Finishes, coatings, construction, and dimensions. - Finishes, coatings, construction, and dimensions for Amplifier Mk 140 shall be strictly in accordance with the applicable drawings, specifications, and publications listed in BUORD LDs 426716, 426717, 426718, 426719, 426721, and 426723.

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3. 2.4 Moisture, fungus, and corrosion. -Amplifier Mk 140 shall be rendered moisture, fungus, and corrosion-resistant in accordance with detail requirements specified in the drawings and specifications listed in BUORD LDs 426716, 426717, 426718, 426719, 426721, and 426723.

3. 2.5 Marking. - Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in BUORD LDs 426716, 426717, 426718, 426719, 426721, and 426723.

3. 2.6 Interchangeability y. - Unless otherwise specified on the applicable drawings, component parts of assembled units fabricated in strict agreement with their applicable drawings shall be interchangeable and shall assemble with mating components of the unit without selection or fitting. If, during assembly of the fabricated parts of a unit, selection or fitting is required and changes in dimensions are necessary, no changes in the parts as delineated on applicable drawings shall be made without prior r e - visions and correction of the applicable drawings (see 3.2. 10).

3. 2.7 Gages. -The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies shall meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, order, or requisition, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractor:s inspection gages. However, such procedures shall not serve to relieve the contractor of his responsibility in the design and manufacture of such gages as may be required in the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3. 2.8 Threads. - Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28 and Supplements thereto. The class of fit for threads shall be as specified on the applicable drawings, and the NOT -GO gaging practice shall be as set forth in the supplement to aforementioned Handbook H28.

3. 2.9 Requests for deviation. - Requests for deviation from this specification, applicable drawings, specifications, publications, materials, and processes specified herein shall be as set forth in OP 400.

3,2.10 Conflicting requirements. - Conflicting requirements arising between this specification or any specification, publication, or drawing listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.3 Definitions

3. 3.1 Vibration. - Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude refers to displacement, which is given in inches peak to peak. The frequency is given in cycles per second (cps).

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3. 3.2 Shock. - Shock is an abrupt change in the motion of a body resulting from the sudden application of a large external force of short duration. Shock, in general, can be described as a build-up of 90 percent of peak force in less than O. 010 second. Shock is described as a pulse in units of gravitational acceleration (g's), duration in milliseconds (ins), and some description of the shock waveform.

3. 3.3 Inclination. - Inclination as referenced in this specification is defined as the acute angle between a given plane and the true vertical plane.

3.3.4 Tropical environment. - Tropical environment is defined as conditions of high humidity, high temperature,. and spore exposure, which present an opportunity for fungi growth.

3.4 Performance requirements and product characteristics. - Amplifier Mk 140 Mods O, 1, and 2 shall meet the following performance requirements and product characteristics:

3.4.1 Performance requirements

3.4.1.1 Servo amplifier component requirements

3. 4.1.1.1 Band-pass filter output to input voltage ratio. - The 400-cycle band-pass filter in each servo amplifier of Amplifier Mk 140 Mods O, 1, and 2 shall have the output to input voltage ratio as specified in 4.5.4.1.

3.4. 1.1.2 Magnetic amplifier balance gain. - The magnetic amplifier of each servo amplifier of Amplifier Mk 140 Mods O, 1, and 2 shall have the input to output characteristics as specified in 4.5.4.2.

3.4. 1.2 Gyro angle order servo amplifier

3.4. 1.2.1 Use. - The gyro angle order servo amplifier shall be usable in either a double- or single-speed synchro transmission system.

3.4. 1.2.1.1 Double-speed normal operation

3.4.1.2.1, 1.1 Gain. - The gyro angle order servo amplifier, connected and tested as specified in 4.5.4.3.1. 1, shall have the following output voltages when the input voltages are as indicated:

Input Voltage (High-Speed Only)	Output Voltage
2 volts	30 ± 15 volts
5 volts	75 ± 15 volts
10 volts	95 ± 2O volts

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Input Voltage Low-Speed Only)	Output Voltage
2 volts	30 ± 15 volts
3 volts	60 ± 2O volts
10 volts	95 ± 2O volts

3. 4.1.2.1.1.2 Zener diode breakdown. - When the gyro angle order servo amplifier is connected and tested as specified in 4.5.4.3-.1.2, the Zener breakdown point shall occur with an input of 2.5 ± 1. 5 volts.

3.4. 1.2. 1. 1.3 Turning eye synchronism indicator. - When the gyro angle order servo amplifier is connected and tested as specified in 4.5.4.3.1.3, the tuning eye synchronism indicator shall be fully open at 0 volt and fully closed at 4.5 * 1.5 volts.

3.4. 1.2.1.1.4 Phase relationship. - When the gyro angle order servo amplifier is connected and tested as specified in 4.5.4.3.1.4, the In-Phase lamp (green) shall go on and stay on when inputs from Test Set Mk 339 are set as follows: (1) Signal Input 2 between 2 and 10 volts and (2) Signal Input 1 between 2 and 10 volts.

3.4. 1.2.1.1.5 Damping circuit. - The gyro angle order servo amplifier, connected and tested as specified in 4.5.4.3.1.5, shall provide the oscilloscope waveforms specified therein.

3.4. 1,2.1.1.6 Noise. - The gyro angle order servo amplifier, connected and tested as specified in 4.5.4.3. 1.6, shall have a noise output of less than 1 volt.

3. 4.1.2.1.2 Single-speed normal operation

3.4. 1.2.1.2.1 Gain. - The gyro angle order servo amplifier, connected and tested as specified in 4. 5.4.3.2.1, shall have the following output voltages when the input voltages are as indicated:

Input Voltage	Output Voltage
0.5 volt	60 ± 15 volts
1 volt	85 *15 volts
2 volts	95 ± 2O volts

3. 4.1.2.1.2.2 Turning eye synchronism indicator. - When the gyro angle order servo amplifier is connected and tested as specified in 4.5.4.3.2.2, the tuning eye synchronism indicator shall be fully open when the input is O volt and fully closed at O. 75 ± O. 25 volt.

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3. 4.1.2.1.2.3 Phase relationship. - When the gyro angle order servo amplifier is connected and tested as specified in 4.5.4.3.2.3, the In-Phase lamp (green) shall go on and stay on when Signal Input 2 from Test Set Mk 339 is set between O. 25 and 10 volts.

3. 4.1.2.1.2.4 Damping circuit. - The gyro angle order servo amplifier, connected and tested as specified in 4.5.4.3.2.4, shall provide the oscilloscope waveforms specified therein.

3,4. 1.2.1.2.5 Noise. -- The gyro angle order servo amplifier, connected and tested as specified in 4.5.4.3.2.5, shall have a noise output of less than 1 volt.

3. 4.1.2.1.3 Double- speed zeroing operation

3. 4.1.2.1.3.1 Gain. - The gyro angle order servo amplifier, connected and tested as specified in 4.5.4.3.3. 1, shall have the following output voltages when the input voltages are as indicated:

Input Voltage (High-Speed Only)	Output Voltage
2 volts	35 volts minimum
3 volts	85 volts minimum

3. 4.1.2.1.3.2 Phase relationship. - When the gyro angle order servo amplifier is connected and tested as specified in 4.5.4.3.3.2, the In-Phase lamp (green) shall go on and stay on when Signal Input 2 from Test Set Mk 339 is set between 2 and 10 volts.

3. 4.1.2.1.3.3 Damping circuit. - The gyro angle order servo amplifier, connected and tested as specified in 4.5.4.3.3.3, shall provide the oscilloscope waveforms specified therein.

3. 4.1.2.1.3.4 Noise. - The gyro angle order servo amplifier, connected and tested as specified in 4.5.4.3.3.4, shall have a noise output of less than 1 volt.

3. 4.1.2.1.4 Single-speed zeroing operation

3. 4.1.2.1.4.1 Gain. - The gyro angle order servo amplifier, connected and tested as specified in 4.5.4.3.4.1, shall have the following output voltages when the input voltages are as indicated:

Input Voltage	Output Voltage
2 volts	25 volts minimum
3 volts	85 volts minimum

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3.4. 1.2.1.4.2 Phase relationship. - When the gyro angle order servo amplifier is connected and tested as 'specified in 4.5.4. 3.4.2, the In-Phase lamp (green) shall go on and stay on when Signal Input 2 from Test Set Mk 339 is set between 2 and 10 volts.

3. 4.1.2.1.4.3 Damping circuit. - The gyro angle order servo amplifier, connected and tested as specified in 4.5.4.3.4.3, shall provide the oscilloscope waveforms specified therein.

3. 4.1.2.1.4.4 Noise. - The gyro angle order servo amplifier, connected and tested as specified in 4.5.4.3.4.4, shall have a noise output of less than 1 volt.

3.4. 1. 2.2 Dielectric. - The wiring of each gyro angle order servo amplifier shall show no evidence of damage during or after the dielectric test specified in 4.5.4.3.5.

3.4. 1. 2.3 Insulation resistance. - The wiring of each gyro angle order servo amplifier shall show a minimum resistance of three megohms and no evidence of damage during or after the test specified in 4.5.4.3.6.

3. 4.1.3 Running depth order servo amplifier

3.4. 1.3.1 Use. - The running depth order servo amplifier shall be usable in either a double- or single-speed (high or low) synchro transmission system.

3. 4.1.3.1.1 Double- speed normal operation

3. 4.1.3.1.1.1 Gain. - The running depth order servo amplifier, connetted and tested as specified in 4.5.4.4. 1. 1, shall have the following output voltages when the input voltages are as indicated:

Input Voltage (High-Speed Only)	Output Voltage
1 volt	25 ± 15 volts
2 volts	(60 ± 15 volts
4 volts	95 *20 volts

Input Voltage (Low-Speed Only)	Output Voltage
8 volts	60 ± 20 volts
9 volts	70 ± 2O volts
10 volts	80 ± 20 volts

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3. 4.1.3.1.1.2 Zener diode breakdown. - When the running depth order servo amplifier is connected and tested as specified in 4.5.4.4.1.2$ the Zener diode breakdown point shall occur with an input of 7.5 ± 2 volts.

3. 4.1.3.1.1.3 Tuning eye synchronism indicator. - When the running depth order servo amplifier is connected and tested as specified in 4.5.4.4.1.3, the tuning eye synchronism indicator shall be fully open at O volt and fully closed at 2.5 * 1 volt.

3.4. 1.3.1.1.4 Phase relationship. - When the running depth order servo amplifier is connected and tested as specified in 4.5.4.4.1.4, the In-Phase lamp (green) shall go on and stay on when inputs from Test Set Mk 339 are set as follows: (1) Signal Input 1 between 8 and 10 volts, and (2) Signal Input 2 between 1.0 and 10 volts.

3.4. 1.3.1.1.5 Damping circuit. - The running depth order servo amplifier, connected and tested "as specified in 4.5.4.4.1.5, shall provide the oscilloscope waveforms specified therein.

3. 4.1.3.1.1.6 Noise. - The running depth order servo amplifier, connected and tested as specified in 4.5.4.4.1.6, shall have a noise output of less than 1 volt.

3.4. 1.3.1.2 Single-speed high normal operation

3.4. 1.3.1.2.1 Gain. - The running depth order servo amplifier, connected and tested as specified in 4.5.4.4.2. 1, shall have the following output voltages when the input voltages are as indicated:

Input Voltage	Output Voltage
1 volt	25 ± 15 volts
2 volts	60 ± 15 volts
4 volts	95 ± 2O volts

3. 4.1.3.1.2.2 Tuning eye synchronism indicator. - When the running depth order servo amplifier is connected and tested as specified in 4.5.4.4.2.2, the tuning eye synchronism indicator shall be fully open when the input is O volt and fully closed at 2.5 1 volt.

3.4. 1.3.1.2.3 Phase relationship. - When the running depth order servo amplifier is connected and tested as specified in 4.5.4.4.2.3, the In-Phase lamp (green) shall go on and stay on when Signal Input 2 from Test Set Mk 339 is set between 1.0 and 10 volts.

3.4. 1.3.1.2.4 Damping circuit. - The running depth order servo amplifier, connected and tested as specified in 4.5.4.4.2.4, shall provide the oscilloscope waveforms specified therein.

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3. 4.1.3.1.2.5 Noise. - The running depth order servo amplifier, connected and tested as specified in 4.5.4.4.2, 5, shall have a noise output of less than 1 volt.

3.4. 1.3.1.3 Single-speed low normal operation

3.4. 1.3.1.3.1 Gain. - The running depth order servo amplifier, connetted and tested as specified in 4.5.4.4.3.1, shall have the following output voltages when the input voltages are as indicated:

Input Voltage	Output Voltage
0.15 volt	40 ± 15 volts
0.30 volt	80 ± 15 volts
0•40 volt	95 ± 20 volts

3.4. 1.3.1.3.2 Tuning eye synchronism indicator. - When the running depth order servo amplifier is connected and tested as specified in 4.5.4.4.3.2, the tuning eye synchronism indicator shall be fully open when the input is O volt and fully closed at O. 35 *O. 15 volt.

3.4. 1.3.1.3.3 Phase relationship. - When the running depth order servo amplifier is connected and tested as specified in 4.5.4.4.3.3, the In-Phase lamp (green) shall go on and stay on when Signal Input 2 from Test Set Mk 339 is set between O. 20 and 10 volts.

3.4. 1.3.1.3.4 Damping circuit, - The running depth order servo amplifier, connected and tested as specified in 4.5.4.4.3.4, shall provide the oscilloscope waveforms specified therein.

3.4. 1.3.1.3.5 Noise. - The running depth order servo amplifier, connected and tested as specified in 4.5.4.4.3.5, shall have a noise output of less than 1 volt.

3.4. 1.3.2 Limiting operation. - The running depth order servo amplifier shall contain a limiting device. When the input data values exceed the range of response of the torpedo being used, a relay shall be energized that shall cut power to the servo amplifier.

3,4. 1.3.2.1 Double-speed limit. - When the running depth order servo amplifier is connected and tested as specified in 4.5.4.4.4, the limiting relay shall be energized and an amplifier output voltage of O volt indicated when the inputs from Test Set Mk 339 are set as follows: (1) Signal Input 2 between O and 10 volts and (2) Signal Input 1 between O and 10 volts.

3.4. 1.3.2.2 Single-speed high limit. - When the running depth order servo amplifier is connected and tested as specified in 4.5.4.4.5, the limiting relay shall be energized and an amplifier output voltage of O volt indicated when Signal Input 2 from Test Set Mk 339 is set between O and 10 volts.

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3. 4.1.3.2.3 Single-speed low limit. - When the running depth order servo amplifier is connected and tested as specified in 4.5.4.4.6, the limiting r clay shall be energized and an amplifier output voltage of O volt indicated -when Signal Input 2 from Test Set Mk 339 is set between O and 10 volts,

3.4. 1.3.3 Dielectric. - The wiring of each running depth order servo amplifier shall show no evidence of damage during or after the dielectric test specified in 4.5.4.4.7.

3.4. 1.3.4 Insulation resistance. - The wiring of each running depth order servo amplifier shall show a minimum resistance of three megohms and no evidence of damage during or after the test specified in 4.5.4.4.8.

3.4. 1.4 Enabling run order servo amplifier

3.4. 1.4.1 Use. - The enabling run order servo amplifier shall be usable in a single-speed synchro transmission system.

3.4. 1.4.1.1 Single-speed normal operation

3.4. 1.4.1.1.1 Gain. - The enabling run order servo amplifier, connected and tested as specified in 4.5.4.5.1.1, shall have the following output voltages when the input voltages are as indicated:

Input Voltage	Output Voltage
0.2 volt	65 ± 2o volts
0.3 volt	80 ± 15 volts
O•6 volt	100 * 15 volts

3.4.1.4.1.1.2 Tuning eye synchronism indicator. - When the enabling run order servo amplifier is connected and tested as specified in 4.5.4.5.1.2, the tuning eye synchronism indicator shall be fully open at O volt and fully closed at 0, 25 *O. 1 volt.

3.4. 1.4.1.1.3 Phase relationship. - When the enabling run order servo amplifier is connected and tested as specified in 4.5.4.5.1.3, the In-Phase lamp (green) shall go on and stay on when Signal Input 2 from Test Set Mk 339 is set between O. 15 and 10 volts.

3. 4.1.4.1.1.4 Damping circuit. - The enabling run order servo amplifier, connected and tested as specified in 4.5.4.5.1.4, shall provide the oscilloscope waveforms specified therein.

3. 4.1.4.1.1,5 Noise. - The enabling run order servo amplifier, connected and tested as specified in 4.5.4.5.1.5, shall have a noise output of less than 1 volt.

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3.4. 1.4.2 Limiting operation. - The enabling run order servo amplifier shall contain a limiting device. When the input data values exceed the range of response of the torpedo being used, a relay shall be energized that shall cut off power to the servo amplifier. When the enabling run servo amplifier is connected and tested as specified in 4.5.4.5.2, the limiting relay shall be energized and an amplifier output voltage of O volt indicated when Signal Input 2 from Test Set Mk 339 is set between O and 10 volts.

3.4. 1.4.3 Dielectric. - The wiring of each enabling run order servo amplifier shall show no evidence of damage during or after the dielectric test specified in 4.5.4.5.3.

3. 4.1.4.4 Insulation resistance. - The wiring of each enabling run order servo amplifier shall show a minimum resistance of three megohms and no evidence of damage during or after the test specified in 4.5.4.5.4.

3.4. 1.5 Amplifier Mk 140 tests

3. 4.1.5.1 Dielectric. - Each amplifier case of Amplifier Mk 140 Mods 0, 1, and 2, connected and tested as specified in 4.5.4.6, shall show no evidence of damage during or after the dielectric test .

3.4. 1.5.2 Insulation resistance. - Each amplifier case of Amplifier Mk 140 Mods 0, 1, and 2, connected and tested as specified in 4.5.4.7, shall show a minimum resistance of three megohms and no evidence of damage during or after the test.

3. 4.1.5.3 Continuity.- Amplifier Mk 140 Mods O, 1, and 2, connected and tested as specified in 4.5.4.8, shall show continuity between the check points specified in paragraph 7c of NAVORD OD 12405.

3. 4.2 Product characteristics

3. 4.2.1 Degree of enclosure. - Amplifier Mk 140 all mods shall be enclosed in a sprayight case in accordance with the requirements specified in MIL-STD-108, Spraytight.

3. 4.2.2 Power requirements. - Amplifier Mk 140 all mods shall re - quire the following power inputs for regular operation:

AC = 115 ± volts, 400 *2 cycles, 3 phase, 300 watts/channel

DC = 24 ± volts

DC = 7 watts max relay load per channel

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3. 4.2.3 Electrical characteristics. - Each of the following servo amplifier components shall conform to the requirements of its applicable drawing:

3.4.2. 3.1 Transformer

Type	BUORD Dwg
Inner stage	1430900
output	1430901
Power	1430902
Tuning Eye	1430906
isolation	1431661

3.4.2. 3.2 Relay. - BUORD Dwg 1148201

3. 4.2.3.3 400-cycle band-pass filter. - BUORD Dwg 1430904

3.4. 2.4 Design," - Amplifier Mk 140 Mods O, 1, and 2 shall have the following design-requirements:

3.4.2.4.1 Test points. - Accessible test points shall be provided for each servo amplifier unit to facilitate checking and testing for input and output 'signals .

3.4.2.4.2 Damping. - Each servo amplifier unit shall use integrated feedback servo damping.

3. 4.2.4.3 Resistors. - The tolerance for all resistors in Amplifier Mk 140 Mods O, 1, and 2 shall be five percent *(5%)unless* otherwise specified.

3. 4.2.4.4 Fuses. - The servo amplifier units of Amplifier Mk 140 Mos O, 1, and 2 shall be individually fused.

3. 4.2.4.5 Construction. - Amplifier Mk 140 Mods O, 1, and 2 shall have the following construction requirements:

3. 4.2.4.5.1 Etched circuits. - Etched circuits shall be used wherever possible to facilitate maintenance and repair, and to minimize space requirements.

3.4. 2.4.5.2 Fuses. - All fuses shall be easily accessible to facilitate replacement.

3.4.2.4, 5.3 Servo amplifier unit replacement. - All servo amplifiers of Amplifier Mk 1.40 Mods 0, 1, and 2 shall be of the plug -in type to facili - tate replacement.

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3. 4.2.4.5.4 Servo amplifier indexing. - All servo amplifier units shall be guide pi indexed to prevent the interchange of servo amplifiers of different functions.

3. 4.2.405.5 Connectors. - Connectors between the *servo* amplifier units and the Amplifier Mk 140 Mods O, 1, and 2 case shall be self-aligning.

3. 4.2.4.5.6 Terminal plate. - The terminal plate of Amplifier Mk 140 Mods O, 1, and 2 shall be gasketed.

3.4. 2.4.5.7 Cover. - The case covers of Amplifier Mk 140 Mods O, 1, and 2 shall be gasketed.

3.4.3 Except as otherwise specified herein, Amplifier Mk 140 Mods O, 1, and 2 shall comply with the requirements specified in MIL-F-18870A

3.5 Environmental requirements

3. 5.1 Vibration. - Amplifier Mk 140 Mods O, 1, and 2 shall withstand vibration requirements as follows:

3. 5.1.1 Vibration, transportation, storage, and handling. - Amplifier Mk 140 Mods O, 1, and 2, packed and packaged as defined in 5.1 and 5.2, shall with stand the tests specified in 4.6.1.1, and thereafter shall show no damage and shall meet all performance requirements and product characteristics of this specification.

3. 5.1.2 Vibration, operating condition. - Amplifier Mk 140 Mods O, 1, and 2, mounted as shown on BUORD Drawings 1485523, 1485524, and 1485525, respectively, shall withstand the tests specified in 4.6.1.2, and during and thereafter shall show no damage and shall meet all performance requirements and product characteristics of this specification.

3. 5.2 Shock. - Amplifier Mk 140 Mods 0, 1, and 2 shall withstand shock requirements as follows:

3. 5.2.1 Shock, transportation, storage, and handling. - Amplifier Mk 140 Mods O, 1, and 2, packed and packaged as defined in 5.1 and 5.2, shall withstand the tests specified in 4. 6.2.1 and thereafter shall show no damage and shall meet all performance requirements and product characteristics of this specification.

3. 5.2.2 Shock, operating condition. - Amplifier Mk 140 Mods O, 1, and 2, mounted as shown on BUORD Drawings 1485523, 1485524, and 1485525, respectively, with circuits energized, shall withstand the tests specified in 4. 6.2.2 and during and thereafter shall show no damage and shall meet all performance requirements and product characteristics of this specification.

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3. 5.3 Inclination. - Amplifier Mk 140 Mods O, 1, and 2, mounted as shown on BUORD Drawings 1485523, 1485524, and 1485525, respectively, shall withstand the inclination tests specified in 4. 6.3 and during and thereafter shall show no damage and shall meet all performance requirements and product characteristics of this specification.

3.5.4 Temperature and humidity. - Amplifier Mk 140 Mods O, 1, and 2 shall withstand the temperature and humidity conditions specified in 4.6.4, and thereafter shall show no corrosion or damage and shall meet all performance requirements and product characteristics of this specification.

3. 5.5 Tropical environment. - Amplifier Mk 140 Mods 0, 1, and 2 shall withstand the fungus resistance tests specified in 4.6.5, and thereafter shall show no fungus growth, corrosion, or other damage, and shall meet all performance requirements and product. characteristics of this specification.

3. 5.6 Marine atmosphere environment. - Amplifier Mk 140 Mods 0, 1, and 2 shall withstand the simulated sea atmosphere conditions and tests specified in 4.6.6, and thereafter shall show no evidence of corrosion or damage and shall. meet all performance requirements and product charac - teristics of this specification.

3. 5.7 Packing, packaging and marking. - The inspector shall astertain that the packing and packaging of. the Amplifiers Mk 140 Mods O, 1, and 2 and their markings conform to Section 5 of this specification.

3. 5.8 Operating life. - Amplifier Mk 140 Mods O, 1, and 2 shall be capable of operating satisfactorily for 3000 hours or a five-year period, whichever occurs first.

3.6 Clean-up. - Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of 100 se, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as rosin flash shall be removed.

3.7 Workmanship. - The equipment, including all parts and accesseries, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking of parts and assembly, painting, machine screw assemblage, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURRANCE PROVISIONS

4.1 Government inspection. - The Government inspector will make such inspections as are necessary to determine that components and assemblies are in accordance with the requirements of the contract,

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pertinent drawings, and specifications. Unless otherwise specified, the se inspections will be conducted in accordance with instructions contained in OP 400.

4. 1.1 Sampling. - Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government inspector in the determination of the acceptability of products submitted by a supplier for Government inspection will, be in accordance with the provisions set forth in MIL-STD-105.

4. 1.2 Pre-production sample. - When specifically required by the contract or order, a representative sample of three (3) Amplifier Mk 140 Mods O, 1, and 2 shall be manufactured using the methods and procedures proposed for the production lot. These shall be submitted to the Government inspector for pre-production tests at an activity designated in the contract or by the Bureau of Ordnance, for purposes of determining compliance with the requirements of the contract, specifications, and drawings. Further production of Amplifier Mk 140 Mods O, 1, and 2 by the contractor prior to approval of the contracting activity or the completion of tests on pre-production samples shall be at the contractor's risk. Pre-production samples accepted shall be applied as part of the quantity specified by the contract or order.

4. 1.3 Lot size. - As applied to Government inspection of units of. product, the term "lot" shall mean “inspection lot; i. e., a collection of units of product submitted by a supplier for Government inspection. Unless otherwise specified, the number of units of product in “inspection lot shall be as determined by the Government inspector and may differ from the quantity designated in the contract or order in a lot for production, shipment or other purpose.

4. 1.4 The government reserves the right to select from each lot, or grand lot of three hundred (300) or less, a random sample of three (3) Amplifiers Mk 140 Mod O, 1, and 2 by the Government Inspector for submission to a Bureau of Ordnance designated activity to determine compliance of the samples with the requirements of the contract, specifications and drawings.

4.2 Contractor's inspection. - Contractor's inspection shall be conducted in accordance with provisions of “Inspection by Contractor" as contained in OP 400.

4.3 Classification of tests. - The inspecting and testing of Amplifier Mk 140 Mods O, 1, and 2 shall be classified as follows:

4. 3.1 Acceptance tests.- The se tests are to be accomplished on each Amplifier Mk 140 Mods O, 1, and 2 being submitted for acceptance under contract. Acceptance tests shall be accomplished under the direction of the Government inspector, who will, at his discretion, either conduct them

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personally or witness their performance by contractor personnel. These tests are defined in 4.5 and shall be conducted in accordance with instructions in NAVORD OD 12405.

4. 3.2 Pre-production and periodic production tests. - Pre-production and periodic production tests are those which are accomplished on samples selected as specified in 4. 1.2 and 4. 1.4 which are representative of the production of Amplifier Mk 140 Mods O, 1, and 2 after the award of the contract to determine that the lot and production meet the requirements of this specification. These tests are detailed in 4.6. Failure of a sample in any of these tests will result in the rejection of the lot or the cessation of production as determined by the procuring activity.

4.4 Test equipment. - The following items of test equipment are required to perform the tests set forth in this specification:

4.4.1 Voltage breakdown tester. - The voltage breakdown tester shall have voltage ranges of 0-3000 volts (rms) AC and 0-4000 volts DC, with maximum currents specified at seven milliamperes AC and five milliamperes DC, respectively.

4.4.2 Megohmer. - The megohmer shall have a test potential of 500 volts DC and ranges of 0-200 megohms, 0-2000 ohms, 0-300 volts DC, and 0-600 volts DC.

4.4.3 Vacuum tube voltmeter. - The vacuum tube voltmeter shall have 12 AC voltage ranges scaling from O-300 volts, reading rms voltages with an internal resistance of 10 megohms. It shall have a full-scale accuracy of 2 percent.

4.4.4 Ohmmeter. - The ohmmeter shall have scales of O-2K, O-200K, and 0-20 megohms with a full scale accuracy of 3 percent.

4. 4.5 Oscilloscope. - The oscilloscope shall have a vertical sensitivity of 1 millivolt to 20 volts per centimeter and a horizontal sensitivity of O. 2 volt to 20 volts per centimeter. It shall have an amplitude calibrator adjustable between O. 2 millivolt to 100 volts with an over-all accuracy of 3 percent.

4.4.6 Frequency checker. - The frequency checker shall have an accuracy of 2 cycles when checking the 400-cycle input.

4.4.7 Special test equipment

4.4.7. 1 Test Set Mk 339 Mod O. - Test Set Mk 339 Mod O, designed to check the operation of the function servo amplifier units, simulates input and output conditions for an electrically set torpedo system, as well as checking input and output phase relationships. The test set shall be designed in accordance with MIL-T-21719.

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4. 4.7.2 Connector shorting strips. - Connector shorting strips shall be made in accordance with figure 1, NAVORD OD 12405. They shall be used to perform the dielectric and insulation resistance tests as specified in paragraph 7, a and b, NAVORD OD 12405.

4. 4.7.3 Connector test cards. - Connector test cards shall be made in accordance with figure 2, NAVORD OD 12405. They shall be used to perform the continuity checks as specified in paragraph 7c, 1, 2, and 3.

4.5 Acceptance tests

4. 5.1 The contractor or Government inspector shall conduct acceptance tests as specified in 4. 3.1 to assure that Amplifier Mk 140 Mods O, 1, and 2 are in compliance with the requirements of 3.2 and 3.4.

4. 5.2 Test conditions. - Unless otherwise specified, Amplifier Mk 140 Mods 0, 1, and 2 shall be subjected to acceptance tests under the following conditions:

4. 5.2.1 Temperature:	Room ambient 65 degrees F. to 95 degrees F. (18. 33 degrees C. to 35 degrees C.)
4. 5.2.2 Altitude:	Normal ground
4. 5.2.3 Vibration:	None
4. 5.2.4 Humidity:	Room ambient to 95 percent relative maximum (except for dielectric tests for which specific humidity is specified)

4. 5.3 Test and inspection equipment and facilities.- The manufacturer shall furnish and maintain all necessary test equipment, facilities, and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity and, when definite requirements are not specified, they shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4. 5.4 Test procedure

4. 5.4.1 Band-pass filter output-to-input voltage ratio. - The 400-cycle band-pass filter of each servo amplifier unit of Amplifier Mk 140 Mods 0, 1, and 2 shall be connected and tested as specified on BUORD Dwg 1430904.

4. 5.4.2 Magnetic amplifier balance gain. - The magnetic amplifier of each servo amplifier of Amplifier Mk 140 Mods 0, 1, and 2, when connected and tested as specified on BUORD Dwg 1287297, shall have the input to output characteristics as specified therein.

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4. 5.4.3 Gyro angle order servo amplifier

4. 5.4.3.1 Double-speed normal operation. - Each gyro angle order servo amplifier, connected and tested as specified in paragraph 4a, NAVORD OD 12405, shall undergo the following double- speed normal operation checks:

4. 5.4.3.1.1 Gain. - Each gyro angle order servo amplifier, connected and tested as specified in paragraph 4a (1) of NAVORD OD 12405, shall show the output voltage gain characteristics specified in 3.4.1.2.1.1.1.

4. 5.4.3.1.2 Zener diode breakdown. - Each gyro angle order servo amplifier shall show the Zener diode breakdown specified in 3.4.1.2.1.1.2, when connected and tested as specified in paragraph 4a (2) of NAVORD OD 12405.

4. 5.4.3.1.3 Tuning eye synchronism indicator. - Each gyro angle order servo amplifier, when connected and tested as specified in paragraph 4a (3) of NAVORD OD 12405, shall operate the tuning eye synchronism indicator as specified in 3.4, 1.2. 1.1.3.

4. 5.4.3.1.4 Phase relationship. - Each gyro angle order servo amplifier shall operate the In-Phase lamps as specified in 3.4.1.2.1.1.4, when connected and tested as specified in paragraph 4a (4) of NAVORD OD 12405.

4. 5.4.3.1.5 Damping circuit. - Each gyro angle order servo amplifier, when connected and tested as specified in paragraph 4a (5) of NAVORD OD 12405, shall give the oscilloscope indications specified therein.

4. 5.4.3.1.6 Noise. - The noise output of each gyro angle order servo amplifier shall not exceed the value specified in 3.4.1.2.1.1.6, when noise-tested as specified in paragraph 4a (6) of NAVORD OD 12405.

4. 5.4.3.2 Single-speed normal operation. - Each gyro angle order servo amplifier, connected and tested as specified in paragraph 4b, NAVORD OD 12405, shall undergo the following single- speed normal checks:

4. 5.4.3.2.1 Gain. - Each gyro angle order servo amplifier, connected and tested as specified in paragraph 4b (1) NAVORD OD 12405, shall show the output voltage gain characteristics as specified in 3.4.1.2.1.2.1.

4. 5.4.3.2.2 Tuning eye synchronism indicator. - Each gyro angle order servo amplifier, when connected and tested as specified in paragraph 4b (2) of NAVORD OD 12405, shall operate the tuning eye synchronism indicator as specified in 3.4.1.2.1.2.2.

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4. 5.4.3.2.3 Phase relationship. - Each gyro angle order servo amplifier shall operate the In-Phase lamps as specified in 3.4.1.2.1.2.3, when connected and tested for phase relationship as specified in paragraph 4b (3) of NAVORD OD 12405.

4. 5.4.3.2.4 Damping circuit. - Each gyro angle order servo amplifier, when connected and tested as specified in paragraph 4b (4) of NAVORD OD 12405, shall give the oscilloscope indications specified therein.

4. 5.4.3.2.5 Noise. - The noise output of each gyro angle order servo amplifier shall not exceed the value specified in 3.4.1.2.1, 2.5, when noise-tested as specified in paragraph 4b (5) of NAVORD OD 12405,

4. 5.4.3.3 Double- speed zeroing operation. - Each gyro angle order servo amplifier, when connected and tested as specified in paragraph 4c of NAVORD OD 12405, shall undergo the following double- speed zero checks:

4. 5.4.3.3.1 Gain. - Each gyro angle order servo amplifier, when connected and tested as specified in paragraph 4c (1) of NAVORD OD 12405, shall show the output voltage gain characteristics as specified in 3.4.1.2.1.3.1.

4. 5.4.3.3.2 Phase relationship. - Each gyro angle order servo amplifier shall operate the In-Phase lamps as specified in 3,4.1.2.1.3.2, when connected and tested as specified in paragraph 4c (2) of NAVORD OD 12405.

4. 5.4.3.3.3 Damping circuit. - Each gyro angle order servo amplifier, when connected and tested as specified in paragraph 4c (3) of NAVORD OD 12405, shall give the oscilloscope indications specified therein.

4. 5.4.3.3.4 Noise. - The noise output of each gyro angle order servo amplifier shall not exceed the value specified in 3.4. 1. Z. 1.3, 4, when noise-tested as specified in paragraph 4c (4) of NAVORD OD 12405.

4. 5.4.3.4 Single-speed zeroing operation. - Each gyro angle order servo amplifier, connected and tested as specified in paragraph 4d of NAVORD OD 12405, shall undergo the following single-speed zero checks:

4. 5.4.3.4.1 Gain, - Each gyro angle order servo amplifier, connected and tested as specified in paragraph 4d (1) of NAVORD OD 12405, shall show the output voltage gain characteristics specified in 3.4.1.2.1.4.1.

4. 5.4.3.4.2 Phase relationship. - Each gyro angle order servo amplifier shall operate the In-Phase lamps as specified in 3.4.1.2.1.4.2, when connected and tested as specified in paragraph 4d (2) of NAVORD OD 12405.

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4. 5.4.3.4.3 Damping circuit. - Each gyro angle order servo amplifier, when connected and tested as specified in paragraph 4d (3) of NAVORD OD 12405, shall give the oscilloscope indications specified therein.

4. 5.4,3.4.4 Noise. - The noise output of each gyro angle order servo amplifier shall not exceed the value as specified in 3.4.1.2.1.4.4, when noise tested as specified in paragraph 4d (4) of NAVORD OD 12405.

4. 5.4.3.5 Dielectric. - Each gyro angle order servo amplifier shall meet the conditions of the dielectric test specified in 3.4.1.2.2, when connected and tested as specified in paragraph 4e of NAVORD OD 12405.

4. 5.4.3.6 Insulation resistance. - Each gyro angle order servo amplifier shall meet the conditions of the insulation resistance test specified in 3.4.1.2.3, when connected and tested as specified in paragraph 4f of NAVORD OD 12405.

4. 5.4.4 Running depth order servo amplifier

4. 5.4.4.1 Double- speed normal operation. - Each running depth order servo amplifier, connected and tested as specified in paragraph 5a of NAVORD OD 12405, shall undergo the following double-speed normal operation checks:

4. 5.4.4.1.1 Gain. - Each running depth order servo amplifier, connected and tested as specified in paragraph 5a (1) of NAVORD OD 12405, shall have the output voltage gain characteristics specified in 3.4.1.3.1.1.1.

4. 5.4.4.1.2 Zener diode breakdown. - Each running depth order servo amplifier shall show the Zener diode breakdown specified in 3.4.1.3.1.1.2, when connected and tested as specified in paragraph 5a (2) of NAVORD OD 12405.

4. 5.4.4.1.3 Tuning eye synchronism indicator. - Each running depth order servo amplifier shall operate the tuning eye synchronism indicator as specified in 3.4. 1.3.1. 1.3, when connected and tested as specified in paragraph 5a (3) of NAVORD OD 12405.

4. 5.4.4.1.4 Phase relationship. - Each running depth order servo amplifier shall operate the In-Phase lamps as specified in 3.4, 1.3.1.1.4, when connected and tested as specified in paragraph 5a (4) of NAVORD OD 12405.

4. 5.4.4.1.5 Damping circuit. - Each running depth order servo amplifier, when connected and tested as specified in paragraph 5a (5) of NAVORD OD 12405, shall give the oscilloscope indications specified therein.

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4. 5.4.4.1.6 Noise. -The noise output of each running depth order servo amplifier shall not exceed the value specified in 3.4. 1.3. 1. 1.6, when noise-tested as specified in paragraph 5a (6) of NAVORD OD 12405.

4. 5.4.4.2 Single-speed high normal operation. - Each running depth order servo amplifier, connected and tested as specified in paragraph 5b of NAVORD OD 12405, shall undergo the following single-speed high normal operation checks:

4. 5.4.4.2.1 Gain. - Each running depth order servo amplifier, connetted and tested as specified in paragraph 5b (1) of NAVORD OD 12405, shall have the output voltage gain characteristics specified in 3.4.1.3.1.2.1.

4. 5.4.4.2.2 Tuning eye synchronism indicator. - Each running depth order servo amplifier shall operate the tuning eye synchronism indicator as specified in 3.4. 1.3. 1.2.2, when connected and tested as specified in paragraph 5b (2) of NAVORD OD 12405.

4. 5.4.4.2.3 Phase relationship. - Each running depth order servo amplifier shall operate the In-Phase lamps as specified in 3.4.1.3.1.2.3, when connected and tested for phase relationship as specified in paragraph 5b (3) of NAVORD OD 12405.

4. 5.4.4.2.4 Damping circuit. - Each running depth order servo amplifier, when connected and tested as specified in paragraph 5b (4) of NAVORD OD 12405, shall give the oscilloscope indications specified therein.

4. 5.4.4.2.5 Noise. - The noise output of each running depth order servo amplifier shall not exceed the value specified in 3.4. 1.3. 1.2.5, when noise-tested as specified in paragraph 5b (5) of NAVORD OD 12405.

4. 5.4.4.3 Single-speed low normal operation. - Each running depth order servo amplifier, connected and tested as specified in paragraph 5c of NAVORD OD 12405, shall undergo the following single-speed low normal operation checks:

4. 5.4.4.3.1 Gain. - Each running depth order servo amplifier, connected and tested as specified in paragraph 5c (1) of NAVORD OD 12405, shall have the output voltage gain characterstics specified in 3.4.1.3.1.3.1.

4. 5.4.4.3.2 Tuning eye synchronism indicator. - Each running depth order servo amplifier shall operate the tuning eye synchronism indicator as specified in 3, 4. 1.3. 1.3.2, when connected and tested as specified in paragraph 5c (2) of NAVORD OD 12405.

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4. 5.4.4.3.3 Phase relationship. - Each running depth order servo amplifier shall operate the In-Phase lamps as specified in 3.4.1.3.1.3.3, when connected and tested for phase relationship as specified in paragraph 5c (3) of NAVORD OD 12405.

4. 5.4.4.3.4 Damping circuit. - Each running depth order servo amplifier, when connected and tested as specified in paragraph 5c (4) of NAVORD OD 12405, shall give the oscilloscope indications specified therein.

4. 5.4.4.3.5 Noise. - The noise output of each running depth order servo amplifier shall not exceed the value specified in 3.4.1.3.1.3.5, when noise tested as specified in paragraph 5c (5) of NAVORD OD 12405.

4. 5.4.4.4 Double-speed limit. - Each running depth order servo amplifier shall show the voltage specified in 3.4.1.3.2.1, when connected and tested as specified in paragraph 5d (1) of NAVORD OD 12405.

4. 5.4.4.5 Single-speed high limit. - Each running depth order servo amplifier shall show the voltage specified in 3.4.1.3.2.2, when connected and tested as specified in paragraph 5e (1) of NAVORD OD 12405.

4. 5.4.4.6 Single-speed low limit. - Each running depth order servo amplifier shall show the voltage specified in 3.4. 1.3.2.3$ when connected and tested as specified in paragraph 5f (1) of NAVORD OD 12405.

4. 5.4.4.7 Dielectric. - Each running depth order servo amplifier shall meet the conditions of the dielectric test specified in 3.4. 1.3.3, when connected and tested as specified in paragraph 5g (1) of NAVORD OD 12405.

4. 5.4.4.8 Insulation resistance. - Each running depth order servo amplifier shall meet the conditions of the insulation resistance test specified in 3.4.1.3.4, when connected and tested as specified in paragraph 5h (1) of NAVORD OD 12405.

4. 5.4.5 Enabling run order servo amplifier

4. 5.4.5.1 Single- speed normal operation. - Each enabling run order servo amplifier, connected and tested as specified in paragraph 6a of NAVORD OD 12405, shall undergo the following single-speed normal operation checks:

4. 5.4.5.1.1 Gain. - Each enabling run order servo amplifier, connected and tested as specified in paragraph 6a (1) of NAVORD OD 12405, shall show the output voltage gain characteristics specified in 3.4.1.4.1.1$_{0}$1.

4.5.4.5.1.2 Tuning eye synchronism indicator. - Each enabling run order servo amplifier shall operate the tuning eye synchronism indicator

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as specified in 3.4. 1.4.1.1.2, when connected and tested as specified in paragraph 6a (2) of NAVORD OD 12405.

4.5.4.5. 1.3 Phase relationship. - Each enabling run order servo amplifier shall operate the In-Phase lamps as specified in 3.4.1.4.1.1.3, when connected and tested for phase relationship as specified in paragraph 6a (3) of NAVORD OD 12405.

4. 5.4.5.1.4 Damping circuit. - Each enabling run order servo amplifier, when connected and tested as specified in paragraph 6a (4) of NAVORD OD 12405, shall give the oscilloscope indications specified therein.

4. 5.4.5.1.5 Noise. - The noise output of each enabling run order servo amplifier shall not exceed the value specified in 3.4.1.4.1. 1.5, when noise-tested as specified in paragraph 6a (5) of NAVORD OD 12405.

4. 5.4.5.2 Single-speed limit. - Each enabling run order servo amplifier shall show the voltage specified in 3.4.1.4.2, when connected and tested as specified in paragraph 6b (1) of NAVORD OD 12405.

4. 5.4.5.3 Dielectric. - Each enabling run order servo amplifier shall meet the conditions of the dielectric test specified in 3.4. 1.4.3, when connetted and tested as specified in paragraph 6c of NAVORD OD 12405.

4. 5.4.5.4 Insulation resistance. - Each enabling run order servo amplifier shall meet the conditions of the insulation resistance test specified in 3.4.1.4.4, when connected and tested as specified in paragraph 6d of NAVORD OD 12405.

4. 5.4.6 Amplifier case dielectric. - Each case of Amplifier Mk 140 Mods O, 1, and 2, connected and tested as specified in paragraph 7a of NAVORD OD 12405, shall pass the dielectric test specified in 3.4.1.5.1.

4. 5.4.7 Amplifier case insulation resistance. - Each case of Amplifier Mk 140 Mods O, 1, and 2, connected and tested as specified in paragraph 7b of NAVORD OD 12405, shall pass the insulation resistance test specified in 3.4.1.5.2.

4. 5.4.8 Continuity. - Each Amplifier Mk 140 Mods O, 1, and 2, connected and tested as specified in paragraph 7c of NAVORD OD 12405, shall show the continuity specified therein,

4.6 Pre-production and periodic production tests

4. 6.1 Vibration

4.6. 1.1 Vibration. transportation, storage, and handling

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4. 6.1.1.1 Mounting. - Amplifier Mk 140 Mods O, 1, and 2, packed and packaged as defined in 5. I and 5.2, shall be secured to the mounting bracket of the testing machine.

4. 6.1.1.2 Orientation. - The package shall be installed in such manner that the direction of vibration will be, in turn, along each of the three rectilinear orientation axes of the package.

4. 6.1.1.3 Test procedure. - The package shall be vibrated in each of the test axes according to Table I.

Table I

Frequency Range (Cps)	Increments of Frequency (Cps)	Amplitude (Double) (inch)
o to 12	1	0.125 ± O. 010
13 to 25	2	0.062 ± O. 008
26 to 55	3	0.031 ± O. 006

4. 6.1.1.4 The package shall be held at resonant frequencies for a minimum of five minutes and shall be held at frequencies 1/2- cycle above and below the resonant frequenciess for a minimum of one minute.

4. 6.1.2 Vibration, operating condition

4. 6.1.2.1 Mounting. - For all tests, the equipment shall be secured to the mounting bracket of the testing machine in the same manner that it shall be secured on board ship.

4. 6.1.2.1.1 Orientation. - The test equipment shall be installed on vibration testing machines in such manner that the direction of vibration will be, in turn, along each of the three rectilinear orientation axes of the equipment as installed on shipboard: vertical, athwart ship, and fore -and-aft. On a horizontal vibration testing machine, the equipment may be turned 90 degrees in the horizontal plane in order to vibrate it in each of the two horizontal orientations. The equipment may be installed in any position.

4,6. 1.2.2 Test procedures. - Each of the tests outlined herein shall be conducted separately in each of the three principal directions of vibration. All tests in one direction shall be completed before proceeding with tests in another direction. The equipment shall be secured to the vibration table as specified in 4.6.1.2.1, and shall be energized to perform its normal functions. All amplitudes of vibration specified herein are average amplitudes of the vibration table. If it is found necessary to repair any damage severe enough to constitute a failure of the equipment, the entire test shall be repeated unless otherwise directed by the bureau or agency concerned.

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4.6. 1. 2.2.1 Exploratory vibration. - To determine the presence of resonances in the equipment under test, the equipment shall be secured to the vibration table and vibrated at frequenciess ranging from 5 cps (or lowest attainable frequency) to 33 cps, at a table vibratory amplitude of O. 010 plus or minus O. 002 inch. The change in frequency shall be made in discrete frequency intervals of 1 cps and maintained at each frequency for about 15 seconds. The frequencies at which resonances occur shall be noted.

4. 6.1.2.2.2 Variable frequency. - The equipment shall be vibrated from 5 cps (or lowest attainable frequency) to 33 cps, in discrete frequency intervals of 1 cps, at the amplitudes shown in Table II. At each integral. frequency, the vibration shall be maintained for five minutes.

4. 6.1.2.2.3 Endurance. - The equipment shall be vibrated for a minimum period of two hours, at the resonant frequencies chosen by the test engineer. If no resonance is observed, this test shall be performed at 33 cps. The amplitude of vibration shall be in accordance with Table 11.

Table II

Frequency Range (Cps)	Table Amplitude (inch)
5 to 15	O. 030 *O. 006
16 to 25	0.020 *o. 004
26 to 33	0.010 *o. 002

4.6.2 Shock

4. 6.2.1 Shock, transportation, storage, and handling. - Amplifier Mk 140 Mods O, 1, and 2, packed and packaged as defined in 5.1 and 5.2, shall be dropped cornerwise from a height of 30 inches onto a solid steel or concrete surface in such a manner that the corner of the package absorbs the full force of the fall. This test shall be repeated until each of the eight corners of the package has received a fall. The height of 30 inches refers to the distance from the steel or concrete surface to the nearest corner of the package when suspended prior to the fall. The package shall be dropped without the use of ropes or any suspending attachments to inhibit the force of the fall.

4. 6.2.2 Shock, operating condition. - Amplifier Mk 140 Mods O, 1, and 2 shall be subjected to the following shock tests:

4. 6.2.2.1 Mounting. - For all tests, the equipment shall be secured to the mounting bracket of the test machine in the same manner that it will be secured on board ship.

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4,6. 2.2.2 Test procedure. - The test shall consist of nine blows being applied: three blows being applied parallel to each of the three principal axes of the equipment. The hammer shall drop from heights of one, three, and five feet. The weight of the hammer shall be 400 pounds. The shock testing machine shall comply with specification MIL-T-17113 (Ships).

4. 6.3 Inclination. - Amplifier Mk 140 Mods O, 1, and 2 shall be subjected to the following inclination tests:

4.6. 3.1 Mounting. - For all tests, the equipment shall be secured to the mounting bracket of the test machine in the same manner that it will be secured on board ship.

4. 6.3.2 Test procedure

4. 6.3.2.1 The equipment shall be inclined at the rate of five to seven cycles per minute in one plane to angles of 45 degrees on either side of the vertical for a period sufficiently long to determine its electrical characteristics under such motion, or for a minimum of 30 minutes.

4.6. 3.2.2 The test specified in 4. 6.3.2.1 shall be repeated with the equipment reoriented 90 degrees to the plane in which it was originally tested.

4.6.3.2.3 At the conclusion of the cyclic tests specified in 4.6.3.2.1 and 4.6.3.2.2, the cyclic motion may be stopped and the inclination adjusted to any angle or angles within 45 degrees of the vertical for a sufficient time to investigate the electrical characterstics of the equipment.

4.6.4 Temperature and humidity. - Amplifier Mk 140 Mods O, 1, and 2 shall be subjected to the following temperature and humidity tests:

4. 6.4.1 Procedure I. - The equipment shall be placed in the test chamber and set up to simulate installed conditions. The chamber temperature shall be between 20° and 38° C. (68" and 100° F.) with uncontrolled humidity. During the first two-hour period, the temperature shall be gradually raised to 71° C. (160°F.). The temperature of 71°C. (160° F.) shall be maintained during the next six-hour period. During the following 16-hour period, the temperature in the chamber shall be gradually reduced to 20° C. (68° F.) and then raised to 38° C. (100° F.) which constitutes one cycle. The relative humidity throughout the cycle shall be 95 percent. The cycle shall be repeated a sufficient number of times to extend the total time of the test to 240 hours (10 cycles). At the conclusion of the 240-hour period, the equipment shall be operated and the data recorded, and a visual examination shall be made for evidence of deterioration or corrosion. Distilled or demineralized water, having a pH value of between 6.5 and 7.5 at 25° C. (77° F.), shall be used to obtain the desired humidity. The velocity of the air throughout the test area shall not exceed 150 feet per minute.

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4.6.4.2 Procedure II. - The equipment shall be placed in the test chamber and set up to simulate installed conditions. The temperature and relative humidity in the chamber shall be 49°C. (120°F.) and 95 percent, respectively. The test conditions shall be maintained for 360 hours. At the conclusion of this period, the equipment shall be operated, the results recorded, and a visual examination shall be made for evidence of deterioration or corrosion.

4.6.5 Fungus resistance tests

4, 6. 5.1 Organisms. - Immediately after Amplifier Mk 140 Mods O, 1, and 2 have been subjected to the temperature and humidity test as specified in 4.6.4, it shall be subjected to the fungi tests specified in paragraphs 4. 4.3.1 through 4.4. 3.3 of MIL-T-945A.

4. 6.6 Marine environment test. - Amplifier Mk 140 Mods O, 1, and 2 shall be subjected to the simulated sea atmosphere test under conditions specified in section 4 of MIL-T-18404. At the conclusion of the test, Amplifier Mk 140 Mods O, 1, and 2 shall meet the performance requirements of this specification.

4. 6.7 operating life. - Amplifier Mk 140 Mods O, 1, and 2 shall be subjected to the following conditioning and accelerated life tests:

4.6.7.1 Conditioning. - In order to establish a reference for the measurement of operating parameters and a valid basis for comparison of test results, Amplifier Mk 140 Mods O, 1, and 2 shall be subjected to the following conditioning process:

4. 6.7.1.1 Normal operation. - The equipment shall first be set up in a room with an ambient temperature of 25 degrees C. plus or minus 5 degrees G. and 50 percent plus or minus 3 percent relative humidity, and operated under normal conditions of power voltage, frequency input, output load conditions, and operating frequency. Following a one and one-half hour warm-up period, gain, zener diode breakdown, damping circuit, and noise plus any specific tests specified herein that are deemed necessary by the Government inspector shall be measured and logged as "reference test data" for comparison with subsequent tests.

4. 6.7.1.2 High temperature operation. - The equipment shall then be set up in a room with an ambient temperature of plus 65° C. which shall be maintained at not less than 90 percent relative humidity. The equipment shall be energized and operated under these conditions at nominal line power voltage, frequency input, output load conditions, and operating frequency for a period of two hours. Near the end of this period and preferably during the last half hour, performance test data shall be taken for comparison with the reference test data (see 4.6.7.1. 1), Minor adjustments to operating controls shall be permitted during this two-hour test run to insure optimum performance.

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4.6. 7.2 Test cycling. - Without shutting down the equipment and immediately following the two-hour, high -temperature conditioning run, the equipment shall be subjected to a series of test cycles specified in 4. 6.7.2.1 through 4.6.7.2.4, while operating in a steady state ambient temperature of between 50° and 55° C. The relative humidity shall be maintained at not less than 90 percent.

4.6. 7.2.1 Test cycling shall be as follows: Increase input power voltage to 110 percent nominal and operate in this condition for a period of one hour and fifteen minutes; then decrease voltage to 90 percent nominal and operate in this condition for one hour. At the end of this period, increase output voltage to nominal and operate for one-half hour, after which time the equipment shall be de-energized for fifteen minutes. During the half -hour period of operation at nominal line input, repeat measurements of operating parameters as specified in 4.6.7.1.2. At the end of the 15-minute shut-down period, energize the equipment and immediately decrease the input voltage to 90 percent nominal; operate in this condition for one hour and fifteen minutes. Decrease voltage to nominal and operate for a period of one-half hour, again repeating measurements specified in 4.6.7. 1.2, followed by a fifteen-minute shut-down of the equipment.

4.6. 7.2.2 The test cycle specified in 4.6, 7.2.1 shall be repeated without interruption for a period of 360 hours (15 complete days).

4. 6.7.2.3 Procedure in case of shut-down. - Should a single shut-down for more than one hour or a total shut-down of eight hours during the 15 days of continuous operation be necessary, the continuation of the test shall be for 15 full days from the time of resumption of the run.

4. 6.7.2.4 Temperature measuring instruments shall be placed at critical points throughout the equipment, covering suspected "hot spot" areas. Data shall be recorded at least once each cycle at the end of the period of high-voltage operation. The log of the data obtained should clearly identify the location of the instrument in the equipment, the temperature, data, and time. This data shall be made part of the report of acceptability y tests.

4,6. 7.2.5 Test data. - Test data accumulated during the accelerated life tests, including details of all failures, shall be provided and included in the report on acceptability tests.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5. 1.1 Level A

5. 1.1.1 Cleaning and drying. - Not applicable

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5.1. 1.[2] Preservative application. - Not applicable.

501. 1.3 Unit packaging. - The amplifiers shall be individually packaged in accordance with Method II of S-pacification MIL-P-116.

5. 1.2 Level B. - Not applicable.

5.103 Level C

5. 1.3.1 Cleaning and drying. - Not applicable.

5. 1.3.2 Preservative application. - Not applicable.

5. 1.3.3 Unit packaging. - The amplifiers shall be individually packaged in accordance with Method III of Specification MIL-P-116.

5.2 Packing

5.2.1 Level A. - In addition to the following, Level A packing shall be in accordance with Specification MIL-P-17555.

5.2. 1.1 Exterior containers. - Each Amplifier Mark 140 Mods O, 1 and 2 shall be packed and boxed in accordance with Bureau of Ordnance Drawings 1485531, 1485532 and 1485533 respectively.

5.2.2 Level B. - Not applicable.

5.2.3 Level C. - Each amplifier packaged in accordance with 5. 1.3.3 shall be packed for domestic shipment in accordance with the manufacturer's commercial practice. The packing shall be adequate to afford protection from deterioration and physical damage during direct shipment from supply source to the first receiving activity for immediate use. Shipping containers shall comply with the carrier rules and r emulations applicable to the mode of transportation.

5.3 Marking

5.3. 1 Special markings. - None, unless otherwise specified.

5. 3.2 Normal markings. - In addition to the markings required by contract or order, unit packages, intermediate packages (when used), and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, including serial numbers.

6. NOTES

6.1 Intended use. - Amplifier Mk 140 is intended for use in under - water and surface fire control systems that utilize electrically set torpe - does.

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6.2 Ordering data. - Procurement documents should specify the specific title, number, and date of this specification and exceptions to this specification, applicable drawings, and other documents. Level C should not be selected unless it is definitely known that the equipment will be used immediately upon receipt.

6.2.1 The attention of the contracting officer is invited to the Quality Assurance Provisions and options in Specification MIL-P-116.

6.2.2 Criteria for use of proper level of preservation, packaging and packing shall be as follows:

For Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For Level B. This level shall be used only when it is definitely known that the item will be held in covered storage in overseas locations for 6 months or less or in domestic locations for an indefinite period.

For Level C. This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity. Level C should not be selected unless it is definitely known that the equipment will be used immediately upon receipt.

Notice: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Governmental procurement operation, the United States Government thereby incurs no responsibility nor any obligation what so ever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or. other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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